

Hash-based MAC

(message authentication code)

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Announcement

- **Midterm exam**

- **Date:** Wednesday, October 21
- **Format:** In class, during our regular class time
- **Cheat sheet:** You may bring one A4-sized cheat sheet. It can be handwritten or typed, and you may use both sides
- **Please mark this date on your calendar now.** There will be no make-up exam unless you have a documented extenuating circumstance
- **Coverage:** The midterm covers everything from the first day of class through October 14
- We will have an exam review session in class on Monday, October 19

Announcement

Homework 2

- **Release:** Wednesday, September 30
- **Due:** Friday, October 9
- **Format:** HW2 will be similar in style to HW1
- **Coverage:**
 - Public Key Cryptography
 - User Authentication
 - Network Access Control
 - Transport Layer Security (TLS)
 - IP Security (IPsec)

Announcement

Course project

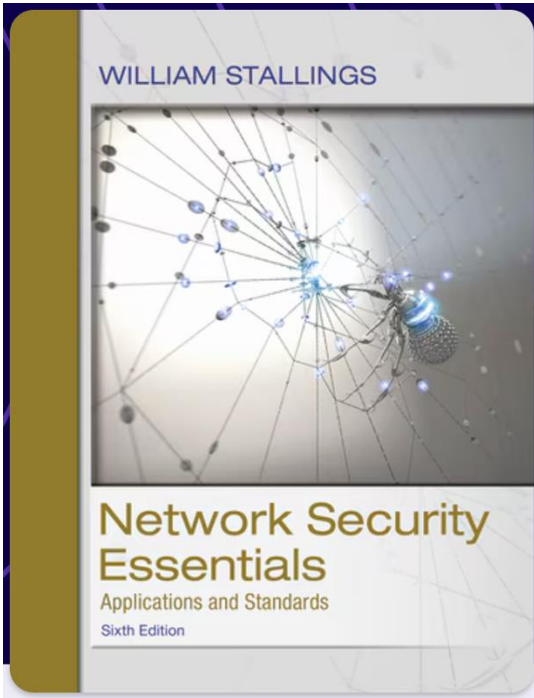
- Please start brainstorming project ideas with your group asap
 - *Choosing a topic.* We **highly encourage** each group to choose its own topic. Anything related to **network security** is fine, and your own interest matters most
 - If you don't have an idea yet, look through network security papers from the **past three years** at conferences such as **IEEE CNS, IEEE GLOBECOM, and IEEE ICC**. Seeing what others are working on can be a great source of inspiration
 - If you still can't come up with a topic, you are welcome to talk with me or the TA, and we can suggest some directions (may not match your interests, may be harder)

Announcement

Course project

- Project proposal. Your proposal should answer the following:
 - **Problem:** What problem are you trying to solve? What impact or benefit would solving it bring, and to whom?
 - **Related work:** What has existing research already achieved on this problem?
 - **Novelty and technical contribution:** What is new about your approach compared with existing work, and what technical contribution do you expect to make?
 - **Team and timeline:** How will your group divide the work, and what is your planned timeline?
- The proposal should be 3 pages, single-column. You may use Word or LaTeX on Overleaf (Overleaf is recommended, template will be provided)

Acknowledgement



The content of slides is from the textbook and Dr. Hu's supplementary materials

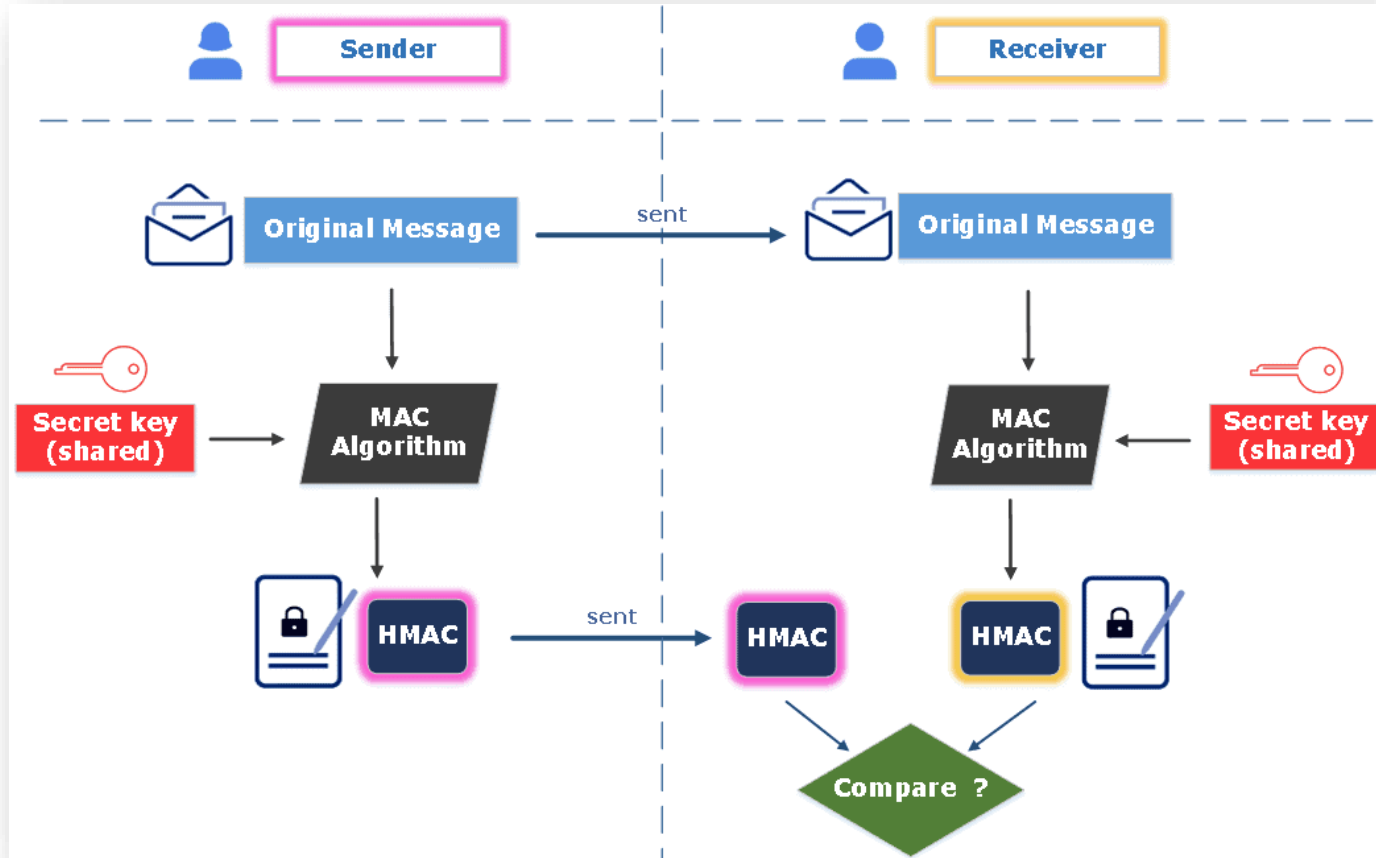
Reminder: Please do not post course materials online or share them with others without the instructor's permission

You can find it on Blackboard

Hash-Based Message Authentication Code

- **HMAC (Hash-Based Message Authentication Code)** is a cryptographic technique that ensures data integrity and authenticity using a hash function and a secret key
 - HMAC ensures that only someone with the specific Secret Key could have generated the code
- The security of HMAC depends primarily on the cryptographic strength of the underlying hash function
- The cryptographic hash function may be MD-5, SHA-1, or SHA-256

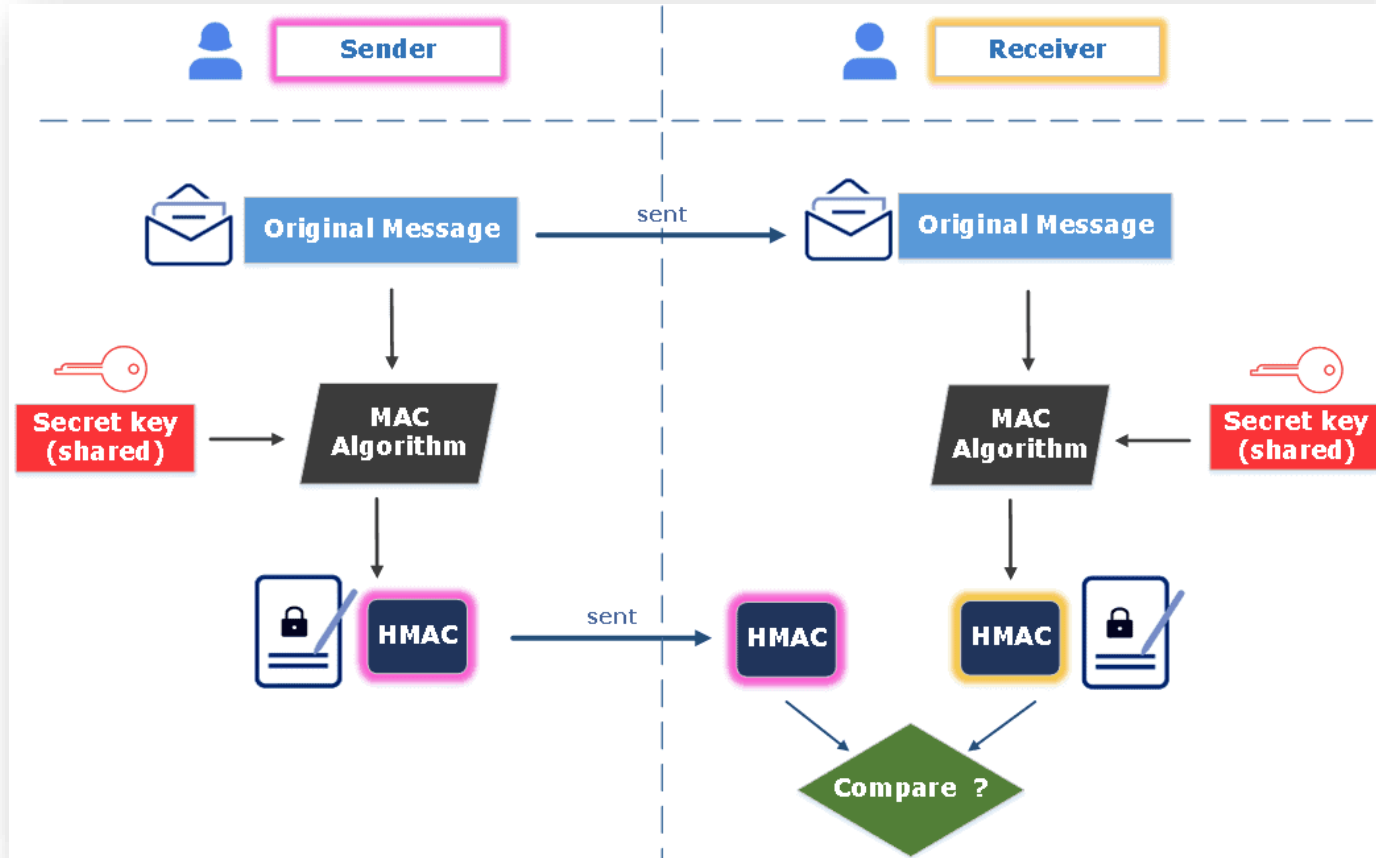
The Workflow of HMAC



At the Sender's Side

- **Input:** The sender takes the **Original Message** and a **Secret Key (shared)**
- **The Algorithm:** Both the message and the key are fed into the **MAC Algorithm** (e.g., SHA-256)
- **The Output:** This produces a unique fixed-length string called the **HMAC**
- **Transmission:** The sender transmits both the **Original Message** and the **HMAC** to the receiver

The Workflow of HMAC



2. At the Receiver's Side

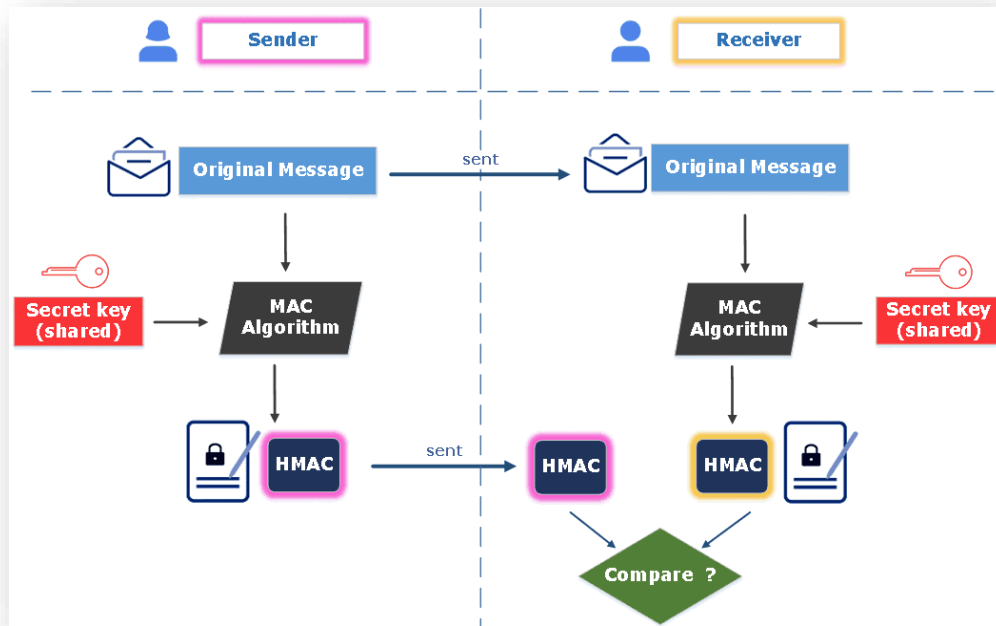
Recalculation: Upon receiving the message, the receiver uses their own copy of the Secret Key (shared) and the received message to run the MAC Algorithm again.

This produces a new HMAC on the receiver's end

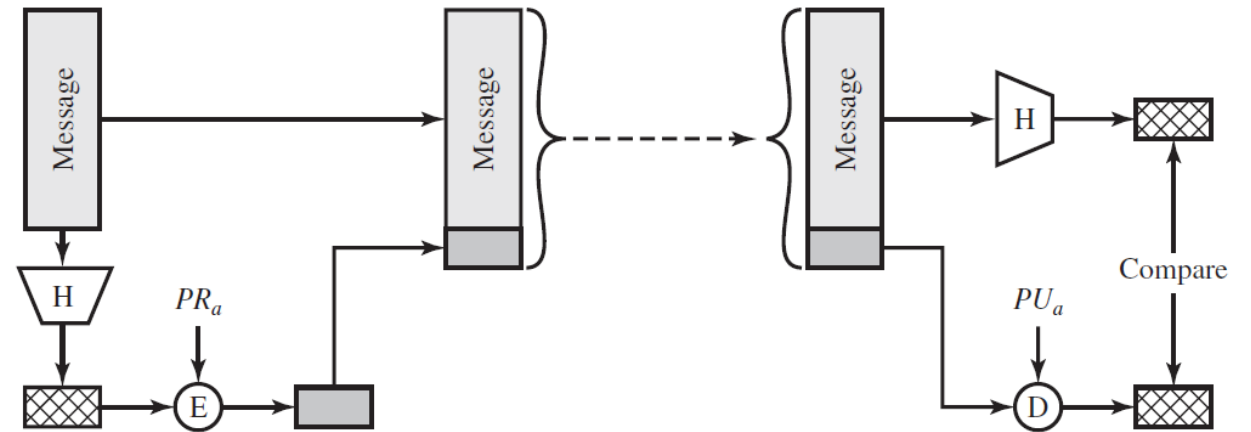
Verification: The receiver compares the HMAC they just calculated with the HMAC they received from the sender

- **If they match:** The receiver is certain the message was not tampered with (Integrity) and that it definitely came from the sender who holds the same secret key (Authenticity).
- **If they don't match:** The message is rejected

The Difference between HMAC and Digital Signature



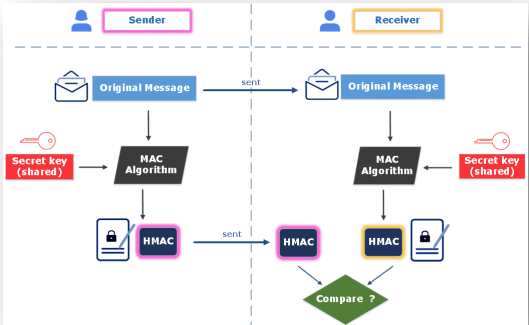
HMAC



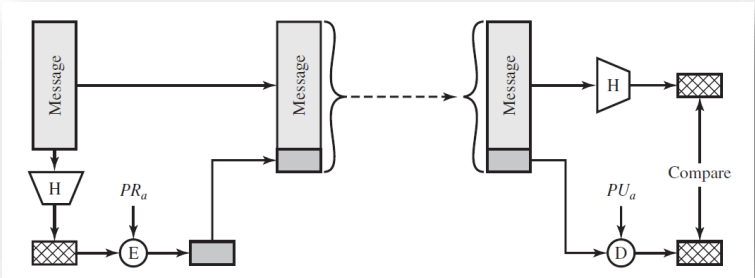
Digital Signature

The Difference between HMAC and Digital Signature

HMAC



Digital Signature



Feature	HMAC	Digital Signature (RSA)
Key Used	Shared Secret Key (Symmetric)	Private/Public Key Pair (Asymmetric)
Performance	Very Fast (Simple hashing)	Slower (Complex modular math)
Non-Repudiation	No. Since both have the key, either could have created the HMAC.	Yes. Only the owner of the Private Key could have created the signature.
Common Use	Fast integrity checks inside an active HTTPS session.	Verifying identity/certificates or signing legal documents.

Secure Hash Algorithm (SHA)

- SHA was originally developed by NIST
- Published as FIPS 180 in 1993
- Was revised in 1995 as **SHA-1**
 - Produces **160-bit** hash values
- NIST issued revised FIPS 180-2 in 2002
 - **SHA-256, SHA-384, SHA-512**
 - Produce 256/384/512-bit hash values
 - Same basic structure as SHA-1 but greater security
- In 2005 NIST announced the intention to phase out approval of SHA-1

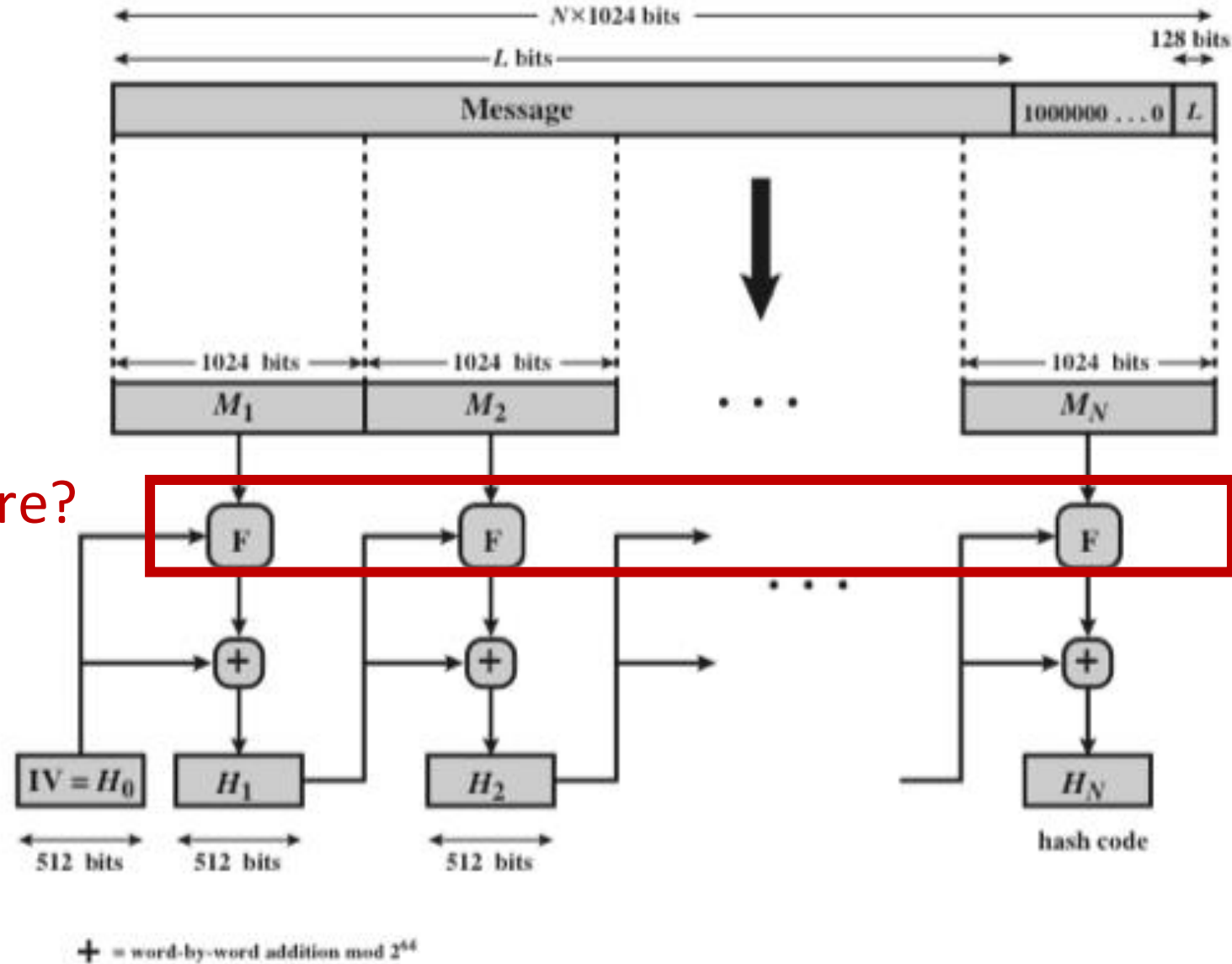
Table 21.1 Comparison of SHA Parameters

	SHA-1	SHA-256	SHA-384	SHA-512
Message digest size	160	256	384	512
Message size	$< 2^{64}$	$< 2^{64}$	$< 2^{128}$	$< 2^{128}$
Block size	512	512	1024	1024
Word size	32	32	64	64
Number of steps	80	64	80	80
Security	80	128	192	256

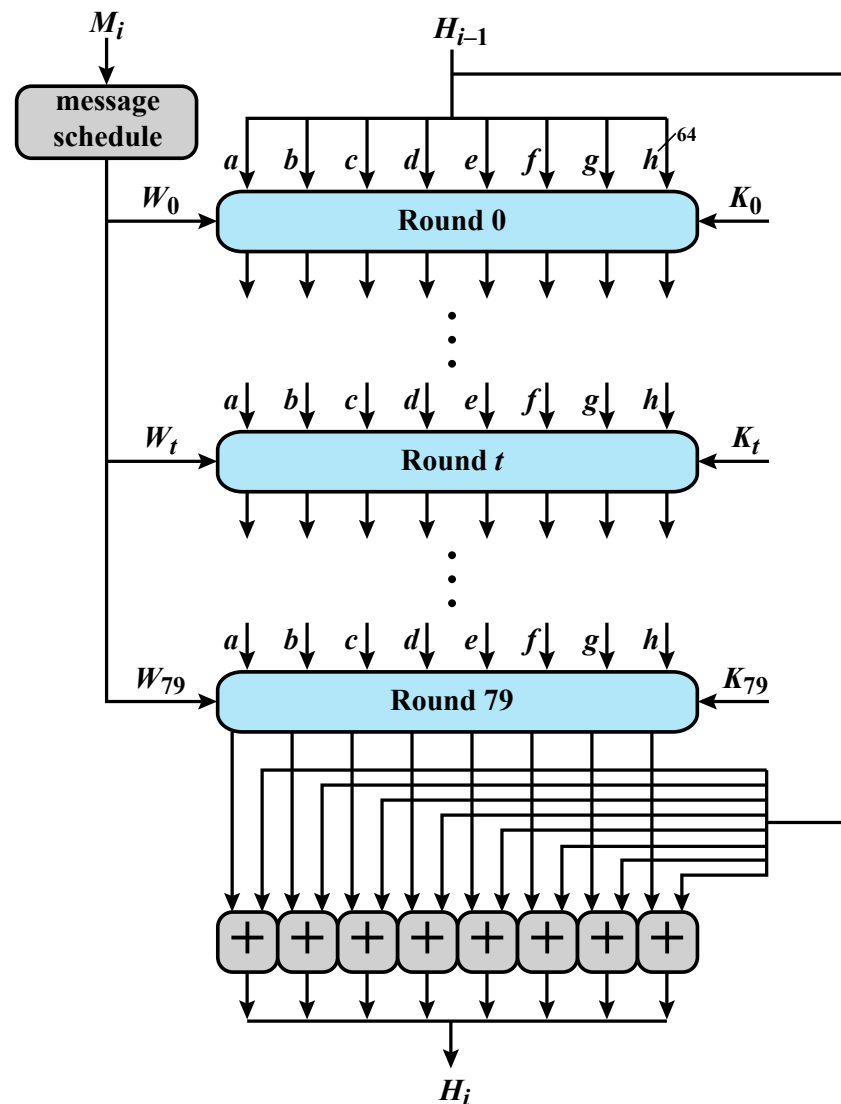
Notes: 1. All sizes are measured in bits.

2. Security refers to the fact that a birthday attack on a message digest of size n produces a collision with a work factor of approximately $2^{n/2}$.

(Optional) Message Digest Generation Using SHA-512



(Optional) SHA-512 Processing of a Single 1024-Bit block



SHA-512 Processing of a Single 1024-Bit Block

To produce M_i , the round function needs

- H_{i-1} (Previous Hash State),
- $W_0 - W_{79}$ (from M_i)
- $K_0 - K_{79}$ (Round Constants) – Each round uses a specific, predefined constant value to introduce further complexity and scramble the data.